

PENGZHAN GUO

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RESEARCH INTERESTS

Parallel Computing, Stochastic Optimization, Machine Learning, Data Mining, and Large-Scale Sequential Decision-Making.

ACADEMIC POSITION

Assistant Professor
Data Science, Duke Kunshan University

July 2022 – Present

EDUCATION

Stony Brook University
Ph.D., Applied Mathematics & Statistics

Nov 2018 – Nov 2021

Stony Brook University
M.S., Applied Mathematics & Statistics

Aug 2017 – Dec 2018

Suzhou University of Science and Technology
B.S., Applied Mathematics and Statistics

Aug 2013 – Jun 2017

RESEARCH GRANTS

Jiangsu Qinglan Project (PI)
Excellent Backbone Young Faculty, 40,000 RMB

2025–27

National Natural Science Foundation of China (Young PI)
Funding: 300,000 RMB

2025–27

Kunshan Supercomputing Center Grant

2022–Present

DKU Zuchongzhi Center Grant (PI)
Funding: 8,000 RMB

2022–23

DKU Start-up Grant

2022–24

AWARDS AND HONORS

Best Oral Paper Award, International Conference on Brain Inspired Cognitive Systems (BICS) *2025*

Best Paper Award, Workshop on Information Technologies and Systems (WITS) *2024*

Excellence in Research in Statistics, Stony Brook University *2022*

Best Paper Award, ACM TMC International Workshop *2021*

RESEARCH PROJECTS

Methodological Contributions in Machine Learning and Data Mining

Weighted Aggregating Stochastic Gradient Descent for Parallel Deep Learning

- Developed a scalable parallel stochastic optimization algorithm under TensorFlow.
- Designed an efficient communication and weighting rule to improve convergence under heterogeneous environments.

Applications in Sequential Decision-Making

Long-Term Career Path Recommendation

- Proposed a reinforcement learning and Markov chain hybrid framework for long-term career planning.
- Demonstrated stable and interpretable performance on large-scale real-world datasets.

Dynamic Taxi Route Recommendation

- Introduced a self-check mechanism within RL to enhance robustness and reliability.
- Integrated deep neural networks to adapt model parameters dynamically to real-time mobility environments.

PUBLICATIONS

Journal and conference rankings follow CORE (A*, A, B, C). An asterisk (*) denotes corresponding author.

Conference Papers

— Under Review —

- **Guo, P.**, Xiao, K., Yang, J. (2024). *LLM-Enhanced Reinforcement Learning with Performance-Aware Adaptive Exploration*. Submitted to ICDM 2026 (CORE A*).

— Published —

- **Guo, P.***, Chen, W. (2025). Similarity-Based Adaptive Stochastic Gradient Descent for Parallel Deep Learning. In *Proceedings of BICS 2025*, Suzhou. (Accepted)
- **Guo, P.**, Xiao, K., Zhu, H., Meng, Q. (2023). Preference-Constrained Career Path Optimization: An Exploration Space-Aware Stochastic Model. In *Proceedings of ICDM 2023*, Shanghai. (CORE A*)
- **Guo, P.**, Ye, Z., Xiao, K. (2019). A Weighted Aggregating SGD for Scalable Parallelization in Deep Learning. In *Proceedings of ICDM 2019*, Beijing. (CORE A*)

Journal Papers

— Under Review —

- **Guo, P.**, Xiao, K., Meng, Q. (2025). Preference-Constrained Career Path Optimization: An Exploration Space-Aware Stochastic Model. Submitted to *IEEE Transactions on Knowledge and Data Engineering (TKDE)*. (CORE A*)
- **Guo, P.**, Ye, Z., Xiao, K. (2025). An Integrated Theory and Solution for the Decision Point Recommendation Challenge. Under review at *IEEE Transactions on Knowledge and Data Engineering (TKDE)*. (CORE A*)
- **Guo, P.**, Xiao, K., Yang, J. (2025). Adaptive Performance-Weighted Multi-Agent Reinforcement Learning for Decision Point Recommendation. Submitted to *Management Science*. (UTD 24)

— Published —

- Wei, Y., **Guo, P.***. (2025). Hybrid Decision Multi-Agent Reinforcement Learning Framework with Dynamical Weighting for Multi-User Sequential Recommendation. *ACM Transactions on Intelligent Systems and Technology (TIST)*. (Major revision)
- **Guo, P.**, Xiao, K. (2024). From Efficiency to Equity: A Multi-User Paradigm in Mobile Route Optimization. *Electronic Commerce Research and Applications*. (Accepted)
- **Guo, P.**, Xiao, K., Ye, Z., Zhu, H., Zhu, W. (2022). Intelligent Career Planning via Stochastic Subsampling Reinforcement Learning. *Scientific Reports*.
- **Guo, P.**, Xiao, K., Ye, Z., Zhu, W. (2021). Route Optimization via Environment-Aware Deep Network and Reinforcement Learning. *ACM Transactions on Intelligent Systems and Technology (TIST)*.
- **Guo, P.**, Ye, Z., Xiao, K., Zhu, W. (2020). Weighted Aggregating Stochastic Gradient Descent for Parallel Deep Learning. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*. (CORE A*)
- **Guo, P.**, Jin, H. (2017). Groemer–Wallen Measure of Asymmetry for Reuleaux Polygons. *Journal of Geometry*.

RESEARCH TALKS

- WITS 2024, Bangkok — *Career Path Recommendation*.
- Summer Workshop on AI for Business 2024, Shanghai — *Career Path Recommendation*.
- Suzhou Youth Mathematicians Workshop 2023 — *Career Path Recommendation*.
- KDD 2023 TMC Workshop — *Preference-Constrained Career Path Recommendation*.
- KDD 2021 TMC Workshop — *Long-Term Career Path Recommendation*.
- Stony Brook University Seminar 2021 — *Intelligent Career Planning*.
- INFORMS 2020 Data Science Workshop — *Career Path Recommendation*.
- ICDM 2019 — *Weighted Aggregating SGD for Parallel Deep Learning*.

PROFESSIONAL ACTIVITIES

Journal Reviewer

- *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*
- *IEEE Transactions on Knowledge and Data Engineering (TKDE)*
- *IEEE Transactions on Parallel and Distributed Systems (TPDS)*
- *IEEE Transactions on Systems, Man, and Cybernetics: Systems*
- *IEEE Transactions on Industrial Informatics (TII)*
- *Electronic Commerce Research and Applications*
- *IEEE Transactions on Big Data*
- *ACM Transactions on Intelligent Systems and Technology (TIST)*
- *Electronic Commerce Research and Applications*
- *IEEE Access*
- *PeerJ Computer Science*

Conference Reviewer

- SIGIR (2022–24)
- KDD (2020–21)
- ICDM (2020–21)
- CIKM (2020–22)
- WSDM (2021)

Conference Program Committee

- ICDM (2026)
- WAIC (2026)